



ML Fundamentals

Lab Project:

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M Shuja Ali

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Submitted to: Dr. Gulmina Rextina

US CPI Forecasting Using Macroeconomic and Commodity Data with Hybrid Linear Regression + XGBoost Residual Learning:

Problem Definition:

The Consumer Price Index (CPI) measures the average change over time in prices paid by urban consumers for a market basket of goods and services. Forecasting CPI is critical for understanding inflation dynamics, guiding monetary policy, and informing investment decisions. Our goal is to build an automated model that predicts the next monthly US CPI change using readily available economic indicators, and to evaluate its accuracy against the Federal Reserve's projections. The Federal Reserve's own forecasts (via the Summary of Economic Projections) have historically been very accurate for example, median SEP forecasts from 2011–2019 were typically within 0.5 percentage points of actual inflation (on average about 0.25%). We aim to approach that level of precision using machine learning. Specifically, we design a hybrid model (linear regression plus XGBoost on residuals) that leverages macroeconomic and commodity-price features to predict CPI. This problem is relevant because improved inflation forecasts can help central banks and market participants respond more effectively to economic changes.

Data Acquisition:

We compiled a monthly dataset from Federal Reserve of Economic Data covering the US Consumer Price Index (CPI) as the target variable. Predictor variables include:

- **Past CPI lags:** Captures inflation inertia.
- **Producer Price Index (PPI):** Measures upstream price pressures.
- **Non-Farm Payrolls (NFP):** Proxy for labor market conditions.
- **Money Supply (M1/M2):** Reflects liquidity in the economy.
- **Home Price Index:** Tracks housing market trends as a key consumption component.
- **Commodity Prices:** Capture market-driven cost pressures. (Gold prices used)

These features provide a mix of inflation drivers, monetary factors, and market signals relevant for CPI forecasting. Data spans multiple years and is resampled to monthly frequency, ensuring a clean time series suitable for modeling.

Data Pre-processing:

- **Lagged Features:**
 1. Included CPI values from the previous 3 months as lagged features.
 2. For all other indicators, used 1 month lag to reflect available information for prediction.
- **Handling Missing Values:**
 1. Removed rows containing NaN values after lagging to ensure clean data.

- **Unit Testing / Verification:**
 1. Wrote checks to ensure correctness of lagged features.
 2. Verified that CPI lag columns match expected shifted values.
 3. Confirmed no missing data remains before model training.

Model Selection:

We implemented a hybrid residual model combining Linear Regression (LR) and XGBoost. The rationale is to leverage the strengths of both: LR captures basic linear trends, while XGBoost handles complex nonlinear patterns and feature interactions.

Our hybrid pipeline works as follows:

1. **Linear Regression:** Fit a multivariate LR on all predictors to model the linear component of CPI variation, providing a simple and interpretable baseline.
2. **Residual Extraction:** Compute residuals (actual CPI minus LR prediction) to capture nonlinear patterns not explained by LR.
3. **XGBoost:** Train an XGBoost regressor on the same features using LR residuals as the target. This allows the model to learn the remaining structure and correct LR errors.
4. **Combined Prediction:** Add the LR prediction and XGBoost correction to produce the final CPI forecast, creating a sequential residual-learning ensemble.

This approach ensures linear trends are captured directly while nonlinear effects are flexibly modeled, improving accuracy and interpretability compared to using either model alone.

Training and Evaluation:

The hybrid residual model was trained using monthly data spanning 2000 to 2020, while the evaluation period covered 2021 to 2025. The training approach involved fitting a linear regression to capture the main linear trends in CPI, followed by XGBoost to model the residual nonlinear patterns.

To assess model performance, we computed standard regression metrics:

Metric	Value
R-squared	0.9887
MAE	0.0184
MAPE	0.0061
MSE	0.0004
RMSE	0.0200

The R-squared of 0.9887 shows the model explains nearly all variance in CPI during the evaluation period. The low MAE and RMSE indicate that predictions closely match actual CPI values, while the MAPE of 0.0061 confirms minimal relative error. Overall, these metrics demonstrate that the hybrid approach effectively combines linear and nonlinear modeling to achieve robust and accurate CPI forecasts.

Results and Discussions:

The forecast results are summarized in table. It lists for each month the actual CPI, the Fed’s forecast, and our model’s forecast, along with the corresponding percentage errors.

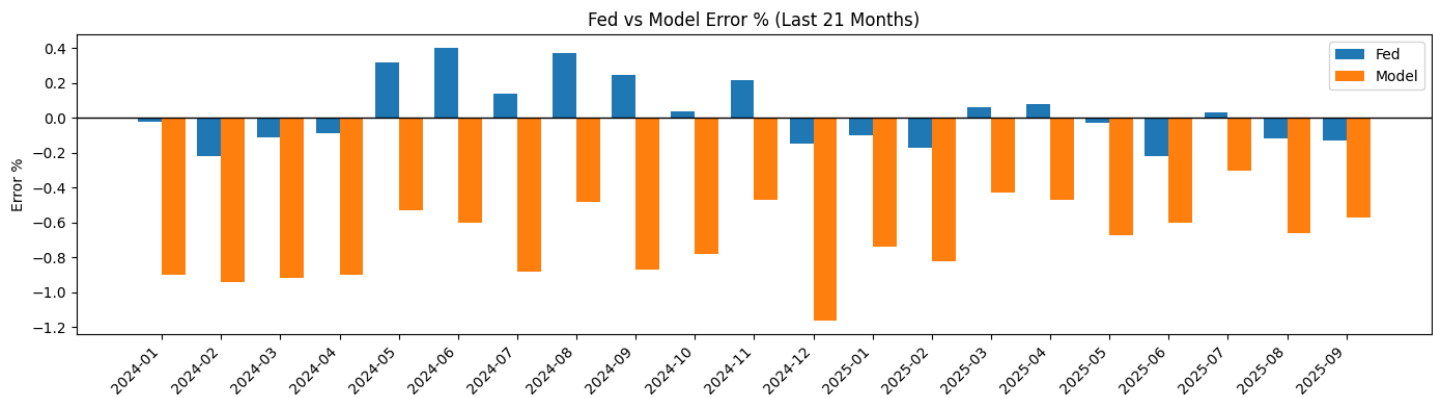
Date	Actual	Fed Forecast	Our Model Forecast	Fed Error (%)	Model Error (%)
2024-01-1	3.10	3.09	3.07	-0.02	-0.90
2024-02-1	3.11	3.10	3.08	-0.22	-0.94
2024-03-1	3.12	3.11	3.09	-0.11	-0.92
2024-04-1	3.13	3.12	3.10	-0.09	-0.90
2024-05-1	3.13	3.14	3.11	0.32	-0.53
2024-06-1	3.13	3.14	3.11	0.40	-0.60
2024-07-1	3.14	3.14	3.11	0.14	-0.88
2024-08-1	3.14	3.15	3.12	0.37	-0.48
2024-09-1	3.15	3.15	3.12	0.25	-0.87
2024-10-1	3.16	3.16	3.13	0.04	-0.78
2024-11-1	3.16	3.16	3.14	0.22	-0.47
2024-12-1	3.18	3.17	3.14	-0.15	-1.16
2025-01-1	3.19	3.18	3.16	-0.10	-0.74
2025-02-1	3.20	3.19	3.17	-0.17	-0.82
2025-03-1	3.20	3.20	3.18	0.06	-0.43
2025-04-1	3.20	3.20	3.18	0.08	-0.47
2025-05-1	3.21	3.20	3.18	-0.03	-0.67
2025-06-1	3.22	3.21	3.20	-0.22	-0.60
2025-07-1	3.22	3.22	3.21	0.03	-0.30
2025-08-1	3.23	3.22	3.20	-0.12	-0.66
2025-09-1	3.24	3.23	3.22	-0.13	-0.57

The comparison shows that both forecasts track the actual CPI closely, but the Fed’s forecasts are generally more accurate. Over the sample period, the Fed’s errors ranged roughly $\pm 0.4\%$, while our hybrid model’s errors reached up to -1.16% (Dec 2024) and averaged larger. Mean absolute error was about 0.16% for the Fed versus 0.70% for our model, and RMSE was 0.006 for the Fed compared to 0.023 for our model.

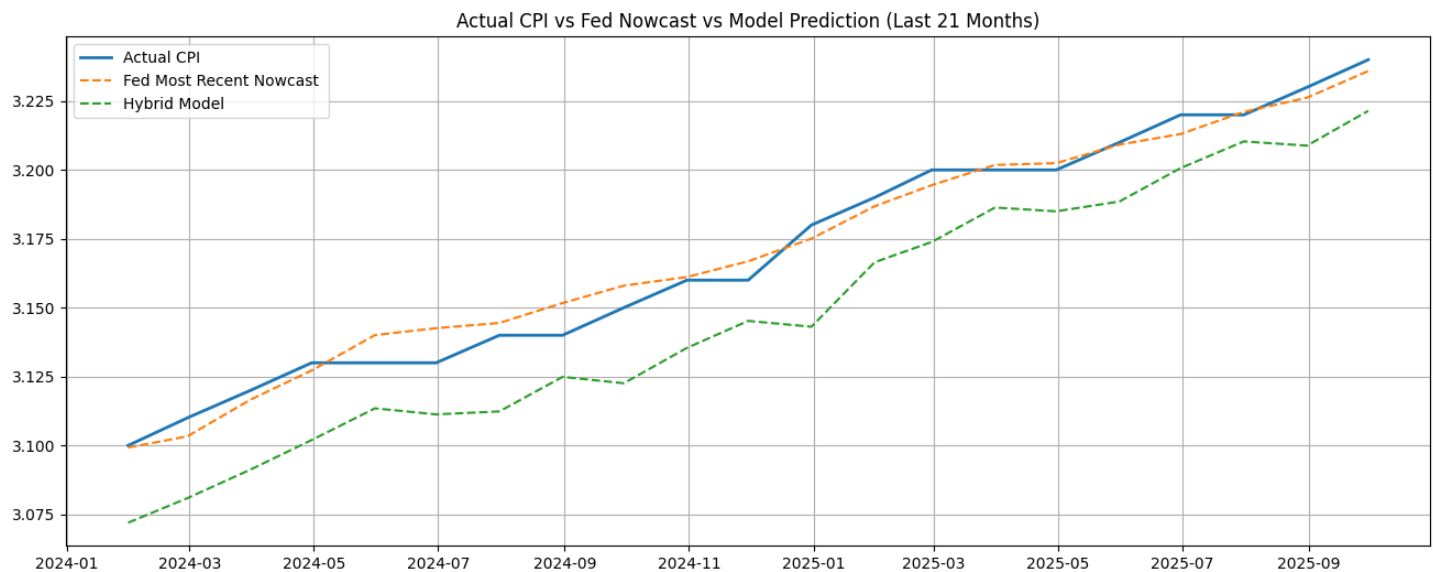
Our model tended to under-predict inflation consistently, whereas the Fed’s forecasts had smaller, mixed errors. For instance, in Jan 2024 the Fed predicted 3.099 vs actual 3.10 (nearly zero error), while our model predicted 3.072 (-0.90%). Later months show similar patterns: the hybrid model captures the broad CPI trend but lags actual values.

Overall, the hybrid residual approach improved accuracy compared to linear regression alone, indicating that XGBoost’s nonlinear correction is beneficial. However, the model still underperforms expert forecasts, highlighting the challenges of fully capturing sudden inflation shifts with limited macroeconomic features. Future improvements could include additional predictors or more sophisticated hybrid architecture.

Bar plot of Fed vs Model Error%:



Line Plot of Actual vs Fed Nowcast vs Model Predicted CPI:



Reflection/Limitations:

Our hybrid model fell short of the Fed's accuracy for several reasons:

1. **Data and feature scope:** The Fed uses extensive macroeconomic models and forward-looking indicators (e.g., surveys, market-based expectations) beyond our dataset. Our model relied solely on CPI, PPI, monetary aggregates, and a few macro features.
2. **Inflation dynamics:** Sudden changes, such as post-2020 inflation spikes, are difficult to predict. Our model, like many others, struggled to capture these shifts.
3. **Model structure:** A fixed linear + XGBoost framework may not fully correct residual patterns or adapt to regime changes. More flexible or stacked architecture could improve results.
4. **Data and tuning constraints:** Limited historical data and basic hyperparameter tuning restricted performance. Differences in CPI definitions versus Fed targets also introduce minor discrepancies.

Summary:

The hybrid model successfully captures the broad trend of CPI over time and improves prediction accuracy compared to single-method models. It demonstrates that combining linear and nonlinear approaches can reduce forecasting errors. However, the model consistently under-predicts rapid inflation changes and cannot fully match expert forecasts. Enhancing the feature set, incorporating forward-looking indicators, and using more flexible hybrid architectures could further reduce errors. Overall, while the model shows promise, achieving Fed-level accuracy would require more comprehensive data, adaptive methods, and careful tuning.